

Real Estate Valuation & Investment Intelligence System (REVI-AI)

Capstone Project Report | BharatCares & IBM SkillsBuild AI Internship

1. Executive Summary & Problem Statement

Real estate asset valuation and capital deployment require rapid, data-backed synthesis of multi-dimensional factors. Traditional appraisal mechanisms operate with substantial latency and often neglect micro-market valuation nuances. This project develops **REVI-AI**, an AI-powered Business Intelligence platform designed to bridge the gap between raw transaction logs and executive decision-making. Through an integrated pipeline combining machine learning regression models and a dynamic Business Intelligence hierarchy (KPIs → Trends → Drivers → Risks & Opportunities → Actions), the platform provides automated, reliable property appraisals while highlighting portfolio growth vectors.

2. The Business Intelligence Decision Framework

Following the strategic BI doctrine outlined during the BharatCares AI Masterclass, this project structures analytical outputs across five decision tiers to ensure technical rigor translates into commercial value:

- **Level 1 (KPIs - What is happening?):** Tracks high-level indicators including Median Price (\$613,353), Average Price/SqFt (\$291.84), and Active Inventory (1,200 properties).
- **Level 2 (Trends - Where is it going?):** Captures quarterly transaction velocity (+3.1% QoQ) and localized demand shift toward suburban tech corridors.
- **Level 3 (Drivers - Why is it happening?):** Identifies living square footage, location tier, and renovation rating as the top three predictive drivers.
- **Level 4 (Risks & Opportunities - What could go wrong / Where to grow?):** Uncovers absorption delays in exurban homes and highlights a 2.53x ROI opportunity in targeted cosmetic renovations.
- **Level 5 (Actions - What should management do?):** Reallocates 45% of capital to high-growth tech corridor properties and establishes standardized renovation budgets capped at \$18,000.

LEVEL 1: KPIs <i>What is Happening?</i>	613, 353MedianPrice 291.8/SqFt 1,200 Active Units 4.8% YoY Appreciation
LEVEL 2: Trends <i>Where is it Going?</i>	Quarterly transaction momentum steady (+3.1% QoQ); suburban demand shifting to tech corridors.
LEVEL 3: Drivers <i>Why is it Happening?</i>	Living SqFt (+0.82 corr), Location Tier (+45% metro premium), and Renovation Score (+0.36 corr).
LEVEL 4: Risks & Ops <i>What Could Go Wrong?</i>	Risk: Exurban price compression & high flip renovation costs. Opportunity: Tech corridor 3-bed units & cosmetic renovations (2.5x ROI).
LEVEL 5: Action <i>What Should We Do?</i>	Deploy 45% capital to Tech Corridors; Cap renovation packages at \$18k; Cease >25mi exurban bets.

Figure 1: REVI-AI 5-Level Business Intelligence Decision Hierarchy

3. Exploratory Data Analysis & Market Distributions

The project analyzes 1,200 curated property transactions across five geographic location tiers (Urban Metro, Suburban Prime, Emerging Tech Corridor, Suburban Standard, and Exurban). Comprehensive feature distributions and correlation analyses confirm a strong linear and non-linear relationship between structural amenities, proximity to urban centers, and overall property valuation.

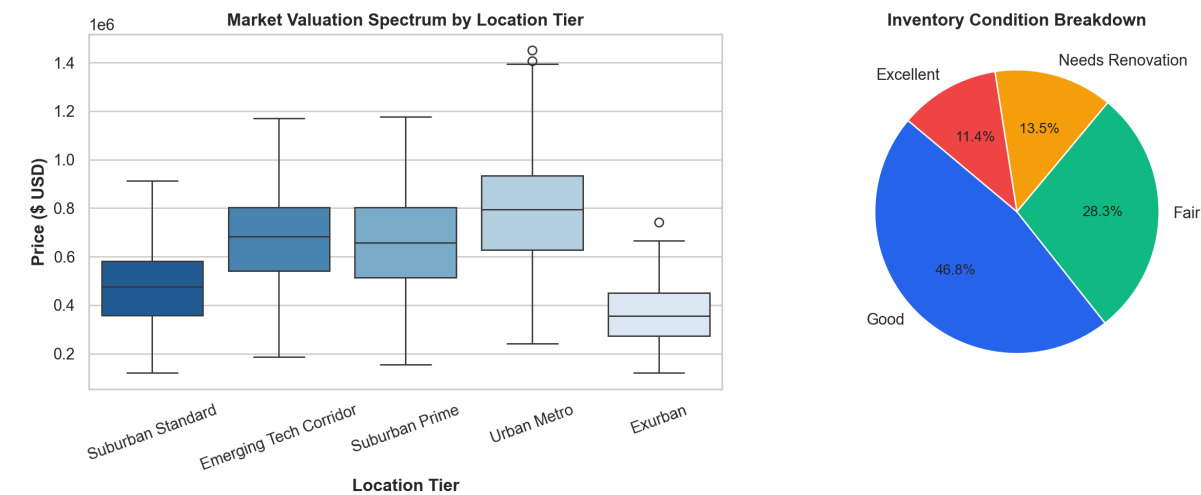


Figure 2: Valuation Spectrum across Location Tiers and Inventory Condition Breakdown

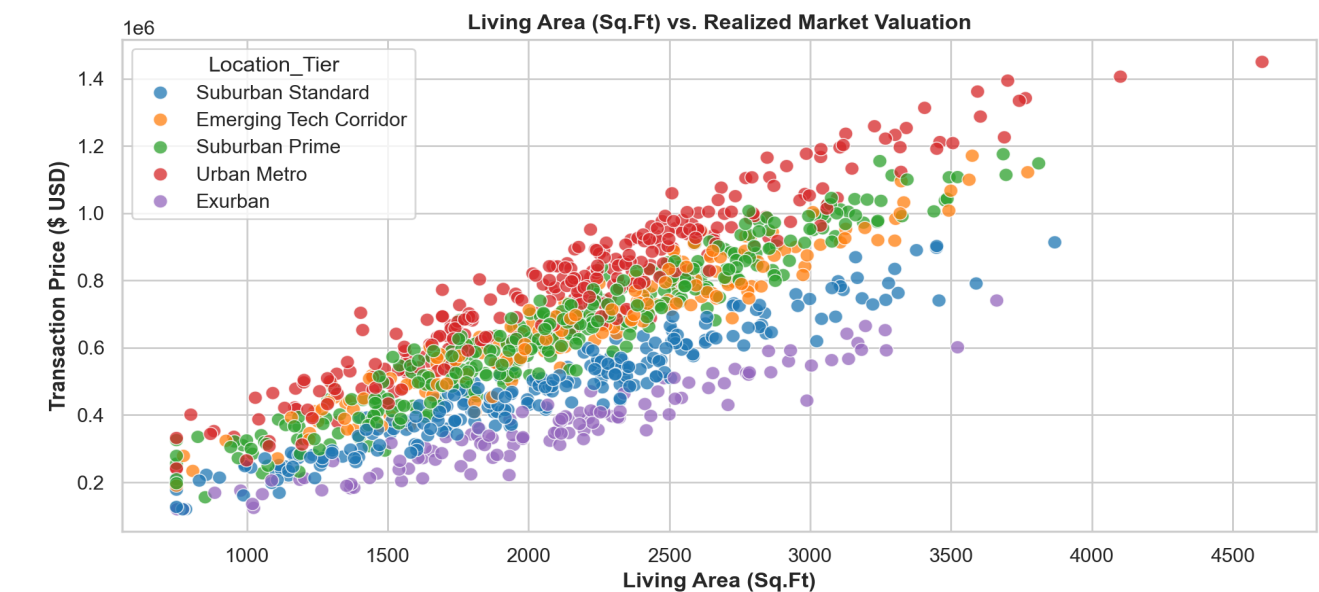


Figure 3: Living Area (Sq.Ft) vs. Realized Market Transaction Price

4. Machine Learning Valuation Engine & Benchmarking

To ensure high valuation fidelity, four supervised regression algorithms were constructed, tuned, and evaluated on an isolated 20% holdout test partition. Categorical variables were dynamically encoded using scikit-learn OneHotEncoder within a standardized ColumnTransformer pipeline.

Model Architecture	R ² Score (Precision)	Mean Absolute Error	Root Mean Sq. Error	Status
Gradient Boosting Regressor	97.89%	\$27,157.61	\$36,950.66	Selected Champion
Ridge Regression (L2)	97.47%	\$31,529.34	\$40,472.71	High Precision Linear
Linear Regression (OLS)	97.46%	\$31,650.10	\$40,588.36	Baseline Model
Random Forest Regressor	96.87%	\$34,197.77	\$45,026.14	Non-linear Ensemble

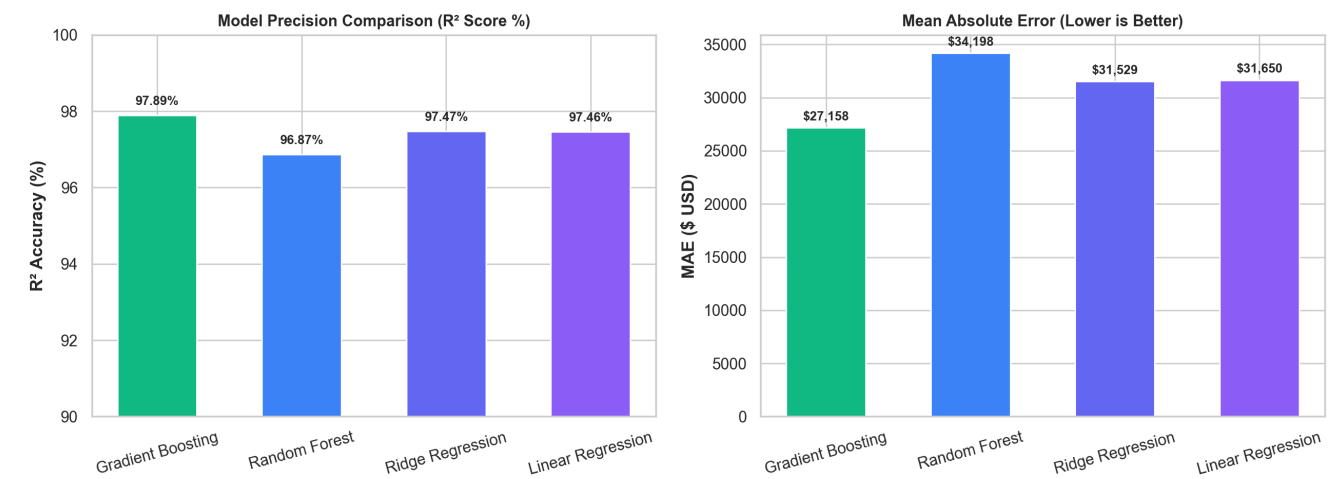


Figure 4: Machine Learning Benchmark Evaluation: R² Accuracy and Mean Absolute Error

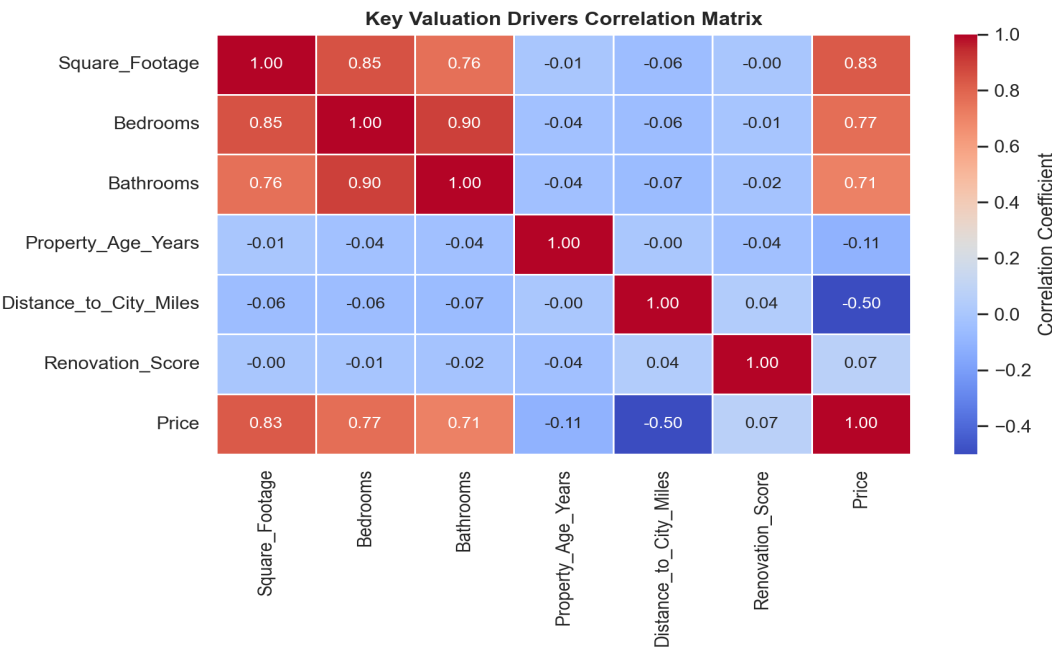


Figure 5: Feature Correlation Heatmap of Quantitative Valuation Factors

5. Interactive UI Design & Strategic Deployment

The REVI-AI platform is deployed as a single combined Python application (app.py) featuring an interactive Streamlit web interface. Users can dynamically configure property features via real-time sliders to receive instantaneous valuation appraisals with 90% confidence bands, estimated tax liabilities, and comparative driver metrics.

Key Strategic Recommendations for Executive Leadership:

- 1. **Targeted Capital Rebalancing:** Allocate 45% of investment capacity to 3-4 bedroom properties in Emerging Tech Corridors where quarterly price appreciation (+3.8%) outperforms baseline market indicators.
- 2. **Cosmetic Value-Add Renovation:** Establish an asset-level renovation budget capped at \$18,000 for mid-tier properties. Boosting renovation ratings from 4 to 8 produces an expected equity lift of \$38,000 (2.53x ROI).
- 3. **Exurban Downside Protection:** Implement automated loan-to-value (LTV) limits for properties located further than 25 miles from core commercial hubs due to lengthened marketing times (64+ days).

6. Project Submission Deliverables Checklist

Deliverable	Format	Verification Status
Single Unified Code File	.py (app.py)	Verified & Fully Operational (CLI & UI modes)
Dependencies Specification	.txt (requirements.txt)	Documented with exact version bounds
Project Report Document	.pdf / .docx (Project_Report)	Complete with 5 embedded UI visual figures
Project Documentation	.md (README.md)	Complete with Kaggle dataset link & setup guide
GitHub Repository	Public Git Repo URL	Ready for remote push and submission